

Logistic Regression

STAT 223 – Applied Analytics

Torin Quinlivan

April 23, 2025



Categorical Response Variables

All of our regression methods have dealt with response variables that were numeric.

Obviously, there are plenty of cases that we want a *categorical* response variable.

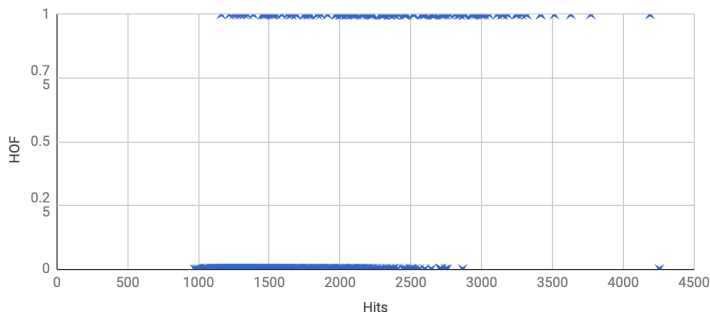
For example, predicting if a candidate will win an election, whether a person will survive a given disease, which team will win a game, etc.

Specifically, we will consider *binary variables* (variables that take a value of 0 or 1), but we'll need to extend our tools to address this.

Latent Variables

What we're going to do is add a layer of abstraction to our regression.

Whereas in linear regression, we directly predict the response variable, to handle a categorical response variable, we will model a *latent variable*, a variable that isn't directly observed and can only be inferred from observations.



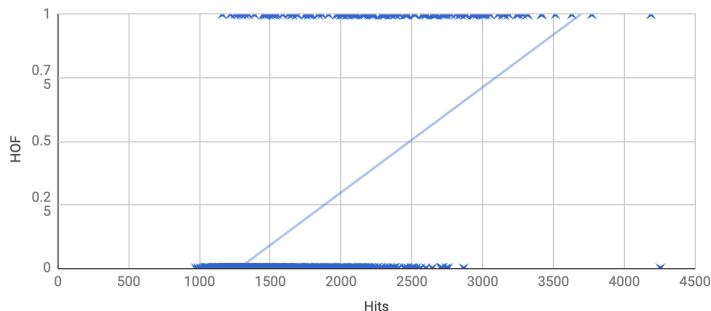
In this case, what we will do is model the observed binary variable Y_i as a BERNOULLI random variable with success probability π_i (our latent variable) and then regress on π_i as a function of our explanatory variables, $\mathbf{X}_i$.

$$Y_i \sim \text{BERNOULLI}(\pi_i)$$
$$\pi_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

Link Functions

However, if you regress on π_i directly, you can end up with the case where you can end up with a predicted probability that could be below zero, or greater than one, which is obviously impossible.

We could just truncate the regression at 0 and 1 if they are less than zero or greater than one, respectively, but clearly that doesn't really make sense.



Link Functions

What we can do is apply a function g to the π_i that we then regress on.

This will allow us to use the tricks of linear regression, but end up with a result that doesn't follow a linear curve.

We call these extensions of a linear model **generalized linear models**. (There's a whole course to be taught here at least, so this is the most surface level of introductions.)

If we use the link function $g(x) = x$, that is called the *identity link*, and gives us basic linear regression.

$$g(y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

Link Functions

Technically, there are as many link functions as there are functions, and different link functions are useful for different scenarios.

The one we will consider is the **logit function**. The logit function is the natural logarithm of the odds of success.

$$y_i = \text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) \implies \pi_i = \frac{e^{y_i}}{1 + e^{y_i}}$$

The inverse of the logit function is sometimes known as the *logistic function* or the *expit function*.

Logistic Regression

Doing regression with a logit link function results in what we call **logistic regression**, which is the most common way to model binary outcomes.

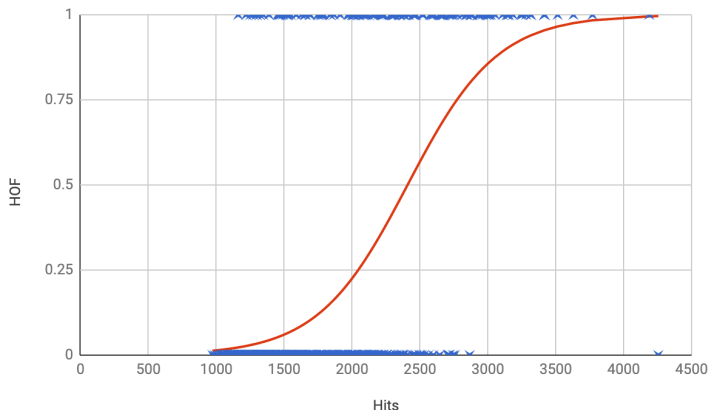
$$Y_i \sim \text{BERNOULLI}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

$$\pi_i = \frac{e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon}}{1 + e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon}}$$

Logistic Regression

Fitting a logistic regression gives us a *sigmoid curve*, a curve that fits entirely within $[0, 1]$ and better models a success probability.



Interpreting Logistic Regression

$$Y_i \sim \text{BERNOULLI}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

$$\pi_i = \frac{e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon}}{1 + e^{\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon}}$$

Obviously, with a link function, interpreting the coefficients of the regression becomes more complicated.

In the case of the logit link, it can be shown that, if the other variables are kept constant and x_i is increased by 1, the success probability π_i will increase or decrease *multiplicatively* by a factor of e^{β_i} .

Interpreting Logistic Regression

